



Improving the Hydraulic Fluid Cleanliness of 794AC Fleet

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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November 2021
3.03-211105-1

1.0 Introduction

Guaranteeing the fluid cleanliness is one of the biggest challenges of the maintenance manager in his quest to guarantee performance of components.

Maintaining the fluid cleanliness below or equal to the Caterpillar guideline/ISO 18/16/13 is a challenge for most of the major mining operations in Brazil even before fluids go into the machine due to oil transport, storage and transferring between tanks and machines. Additional factors can affect the cleanliness inside of the machines' hydraulic systems due to leaks in damaged hoses, broken seals, and overall unintentional contamination during maintenance; in conditions of high utilization where maintenance has to be done in windows of opportunity, which may occur outside of a clean bay, contamination control becomes more and more important but also more and more difficult resulting in the need for filtration.

One of Sotreq's largest customers encountered the following SAE 10W SOS results out of specification in the hydraulic systems of their 794AC Fleet and bulk storage tanks.

Cleanliness the customer's 10W	ISO
jan/21	23/21/18
fev/21	22/20/16
mar/21	21/19/16
abr/21	22/20/18
mai/21	21/20/15
jun/21	20/18/15
jul/21	22/20/17
ago/21	21/18/12
set/21	23/21/18



Source/Picture 1: Customer/Dealer SOS Results and SAE10W at oil tank in site

This customer already has an oil filtration system for the bulk storage tanks, so they requested the support of the dealer to find a solution to maintain oil cleanliness on board the trucks. This type of solution would guarantee clean oil throughout the machine, independent of existing conditions, and result in improved component life and reduced downtime during PM's (preventive maintenance). With this solution the customer is also looking for improved serviceability of the electric drive trucks, lower number of tools used in the overall maintenance and an overall lower maintenance ratio. Their vision was to apply the principles of conventional off-board kidney looping machine (filtration/dialysis, participle count, etc.) typically used during PMs to an on-board kidney looping system

Sotreq developed an on-board continuous filtration (oil dialysis) system. This aftermarket solution aligns with market practices of continuous cleaning for hydraulic compartments for mobile equipment.

Implementation of this system in the 794AC trucks has shown good results – clean oil, clean compartments, and lower maintenance ratio. Longer life for the hydraulic components is the ultimate goal for this system; it is expected that this will be the ultimate result of the developed system.

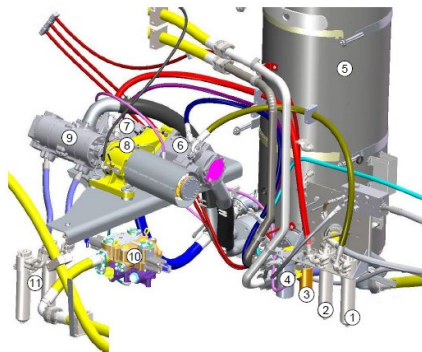
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This on-board solution can be readily implemented by other dealers with similar customer requests/needs. (Notes: Caterpillar does not have an on-board solution. There is no expectation of Caterpillar developing an on-board system. This best practice demonstrates a dealer-based solution.)

2.0 Best Practice Description

The 794AC hydraulic system of the 794AC trucks is comprised of approximately 11 principal items, with filters at the entrance of each component capable of filtering particles of 4 microns, as shown in the figure below. As a note, 10W oil is used throughout the Brazilian market.

The customer requested that Sotreq come up with a “bulletproof” solution for keeping the hydraulic system clean that didn’t adversely affect the components or overall functionality and reliability of the truck. The challenge was to transform a conventional off board dialysis system to a system integrated (onboard) with the OEM equipment (base machine).



Picture 2 — Steering Filter, (2) Radiator Fan Filter, (3) PTO filter, (4) Hydraulic Tank Filter, (5) Hydraulic Tank, (6) Steering Tank, (7) Radiator Fan Tank, (8) PTO, (9) Lift Tank, (10) GP Lift Valve, (11) Lift Filters.

As seen in the Best Practice 1.01.210827-1, this Brazilian customer operates a fleet of 794AC trucks with maintenance based on the lean concept with focus on improving the productivity of the PM. To do conventional dialysis in the shop demands preparation time, connection checks, regular maintenance and testing of the dialysis machines and allocation of personnel/time for the actual dialysis. Conventional analysis does not support improved PM productivity and is not 100% effective in the long term of operation.

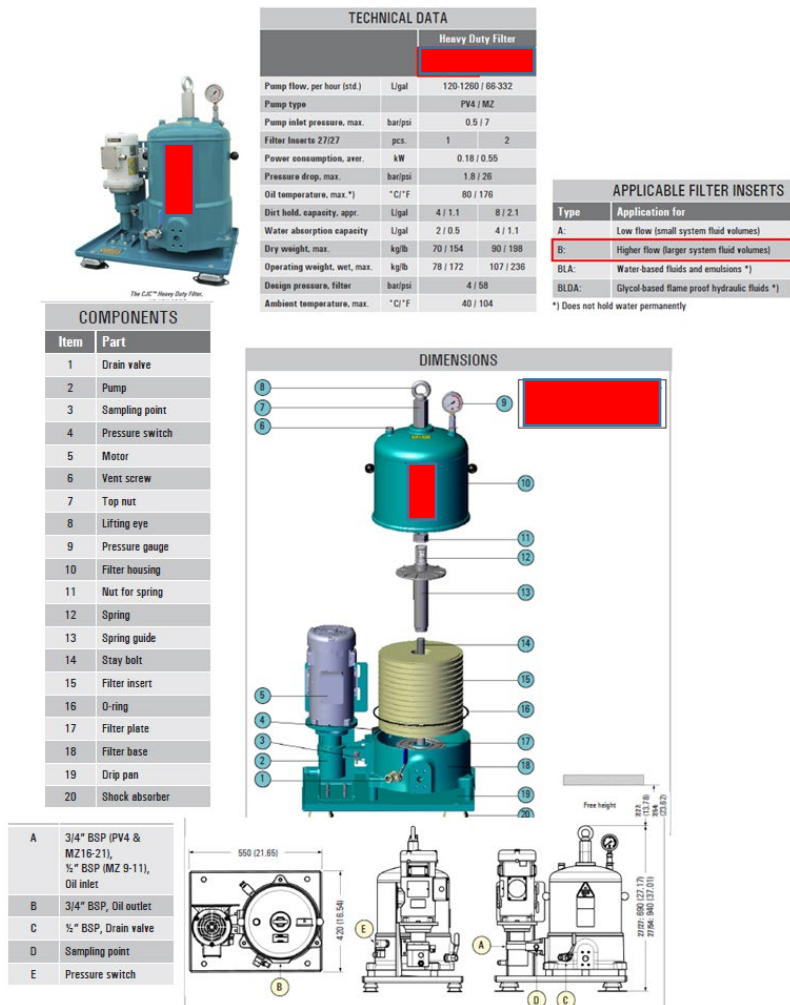
Sotreq evaluated OEM dialysis systems in its existing Sotreq network and partnered with an OEM, that they had previously worked with on other applications, to create an on-board filtration system to 1) deliver high quality compartment cleaning, 2) conform to the recommended Caterpillar ISO specification, 3) facilitate quality and efficient maintenance and 4) cause no risks to the machine (as it is connected directly to the machine).

A filtration unit with 1.458 L of flow and volume for the hydraulic system of the 794AC at 3 times per 12 working hour shift was chosen. It was sized to hold 4 Kg of dirt, 2 L of water and 4 Kg of oil oxidation. It had the capability to filter 98.7% of the particles of 3 microns or above and 50% of the particles between 0.8 to 3 microns.

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This on-board dialysis system complements the filtration system of the 794AC. It provides additional protection and redundancy in situations 1) where the machine hydraulic system has a bypass due to high contamination and 2) on some 794AC trucks, there are no electronic restriction indications/alarms since in most cases there are no sensors that are ultimately connected to the machine monitoring system. This system has the potential to be a powerful tool for the long-term extension (above 4,000 hours) of the fleet's hydraulic oil (oil oxidation retention).

NOTE: EACH CUSTOMER OR DEALER CAN DEFINE THE BEST ON-BOARD FILTRATION UNIT FOR THE FLEET WHILE RESPECTING THE TECHNICAL ASPECTS AND RISK FOR THE EQUIPMENT. THIS BEST PRACTICE DOES NOT AIM TO PRODUCE A COMMERCIAL/FOR SALE SYSTEM OR TO TRANSFORM IT INTO A PRODUCTION SOLUTION. THE BEST PRACTICE DEMONSTRATES RESULTS OF A FIELD TEST PERFORMED FOR THIS CONCEPT.



Picture 3: On-board filtration unit selected

The basic design of this on-board solution follows the principals of the conventional dialysis with oil suctioned from the hydraulic tank, passed through the filtration unit (now onboard vs. off board) and then returned to the same hydraulic tank. A clean oil supply at all time is the result.

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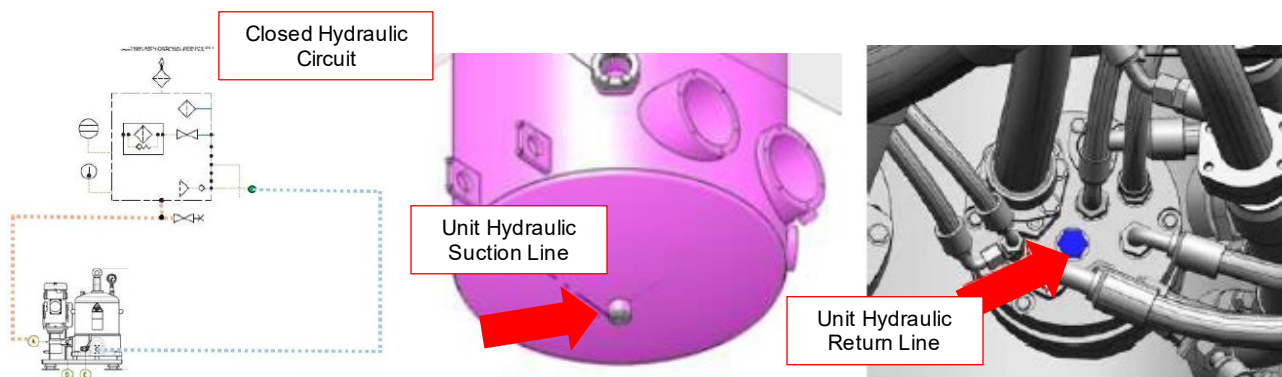
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Note: Caterpillar genuine parts are used for the hoses, connectors, clamps, and other parts to support the configuration and installation of the on-board filtering system. The filtration unit itself is the only non-Caterpillar part, which is available from an OEM that Sotreq has previously partnered with in applications for mining, construction, industrial and power generation.

Note: If the on-board filtration system has any failure in the unit or a connection between the unit/installation, the system will not damage the truck hydraulic system.

- If the on-board filtration system has any restrictions in the hydraulic line, the unit discharges the oil into the tank without any damage since it is not connected to the main hydraulic line.
- There are no restriction valves installed throughout the unit.
- If there is an electrical failure no damage will occur as the unit hydraulic lines work separately from the main truck hydraulic lines.
- The assumption is the on-board filter connection is the same as connecting an off-board system.



Pictures 4: On-board filtration Hydraulic Flow, Suction Line and Return Line

3.0 Implementation Steps

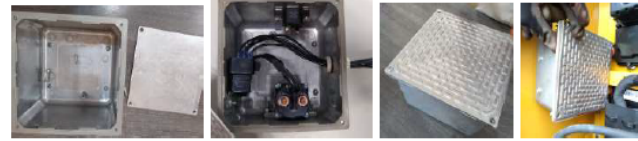
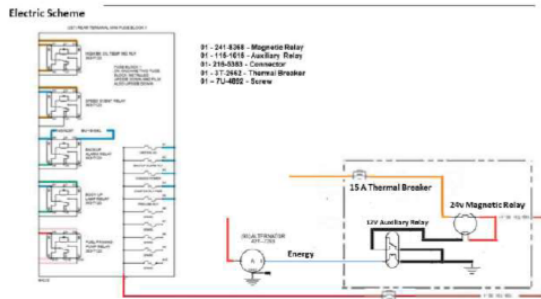
The project was developed around three fronts of implementation (once the filtration unit was chosen by on the fleet's characteristics):

- 1) Installation of the filtration system power supply.
- 2) Installation of the filtration system unit on board the truck and attached to the platform
- 3) Training of the mechanics on both operation and maintenance

STEP 1: Installation of the filtration system power supply

The power system is shown below. An electrical box with thermocouples was located next to the ECM cabinet. The installation took into consideration existing safety guidelines for mounting and connection of common accessories.

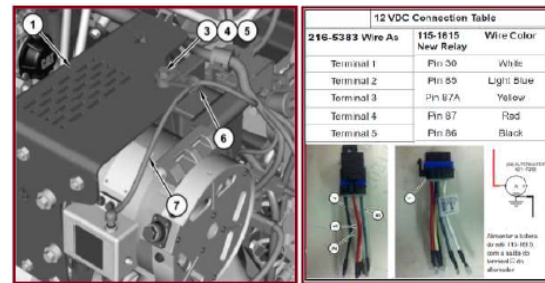
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- 1- Install and fix the electrical box installed in the equipment's ECM's cabinet.
- 2- Assemble the electrical circuit according to the diagram above, using the following components: Auxiliary relay (115-1615), magnetic relay (241-8368) and 15A thermal breaker (3T-2662) inside the passage box - Aluminum.
- 3- Use the harness (127-4872) coming out of the alternator's R terminal, install a type terminal (female bullet) at the end of the wire and supply the 12V auxiliary relay coil to the light blue connector wire, pin 85 of the relay.
- 4- Use plastic clamps (7K-1181) to route the harness (127-4872 single wire) next to the machine harness, from the changer to the 15x15 pass box - aluminum.

Necessary parts for electrical system

Type	Pn	Qty	Description
Electric	9R-2013	4	BOSS
Electric	8T-4137	4	Screw
Electric	8T-4121	4	Washer
Electric	241-8368	1	Magnetic Relay
Electric	115-1615	1	Assistant Relay
Electric	3T-2662	1	Thermal Breaker
Electric	7K-1181	20	Plastic Clamp
Electric	127-4872	1	Whip
Electric	7U-4962	2	Screw (8-32x0.75in)
Electric	6V-0683	2	Grommet
Electric	NA	5 m	PP Cable 2x2.5
Electric	6V-0560	1	Ring Connections
Electric	***	2	Tubular Terminal
Electric	Electric Box	1	Junction Box 15x15 - Aluminum

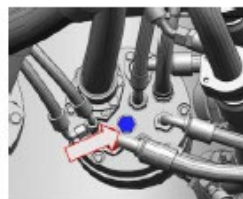
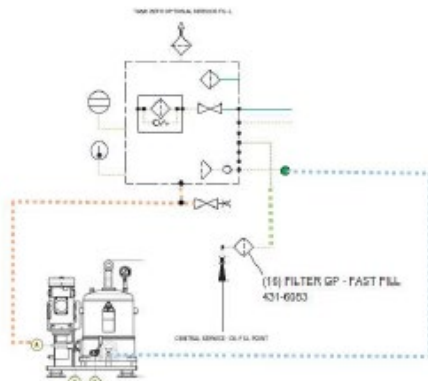


Up arrow shows connection of alternator R terminal, 12V output

Pictures 4 – Installation guideline to step 1

Step 2: Installation of the filtration system unit on board the truck attached to the platform.

This part of the installation pertains to the hydraulic connections (from suction to return-to-tank) and the filtration unit itself. The suction lines are located below the hydraulic tank and directed to the filtration unit (red). After filtering, the oil flows through the hoses (blue) with access to the hydraulic tank through the hydraulic line manifold.



Type	Pn	Qty	Description
Suction	456-8430	08 m	Bulk Hose (Cut as needed).
Suction	360-3697	1	Plug
Suction	238-5084	4	Seal
Suction	287-7301	1	Register
Suction	448-9011	1	"T" Adapter
Suction	8T-0197	1	Reduction
Suction	124-2153	2	Hose Terminal
Suction	6V-0746	2	Hose Ring
Suction	086-2746	1	Reduction
Suction	6V-0408	1	Reduction
Suction	228-7092	2	Reduction

One-Wire Braid Hose ToughGuard (716TG)

Part Number	HOSE I.D. (mm - in)	HOSE O.D. (mm - in)	HOSE SIZE	MIN. BURST PRESS. (MPa - psi)	MAX. WORK PRESS. (MPa - psi)	MIN. BEND RADIUS (mm - in)
456-0400	10.1 - .75	30.2 - 1.19	1/2	42.0 - 600	10.5 - 150	240 - 9.45

Heavy Duty Straight Thread O-Ring - Hex Head Plugs

Part Number	Thread Size (in. - FPI)	O-Ring Seal
360-3697	1.5/16-12	150-0405

STOR Port / STOR Port / Male STOR - Tee

Part Number	Thread size	L1 Length	Thread size	L2 Length	Thread size	L3 Length	Material
448-9011	1.5/16-12	33.0	1.5/16-12	33.0	1.5/16-12	51.15	Steel

Male ORFS / Male STOR

Part Number	Thread Size (in.)	Thread Size (in.)	HEX Size	Length (in.)	Material
01-0192	1.5/16-12	1.5/16-12	36.1	82.5	Steel

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Type	Pn	Qty	Description
Suction	456-8430	08 m	Bulk Hose (Cut as needed)
Suction	360-3697	1	Plug
Suction	239-6284	4	Seal
Suction	287-7321	1	Register
Suction	448-9011	1	T" Adapter
Suction	87-0197	1	Reduction
Suction	124-2153	2	Hose Terminal
Suction	6V-9746	2	Hose Ring
Suction	086-2746	1	Reduction
Suction	6V-0400	1	Reduction
Suction	228-7052	2	Reduction

One-Wire Braid Hose ToughGuard (716TG)

Part Number	HOSE I.D. (mm - in)	HOSE O.D. (mm - in)	HOSE LENGTH (m - ft)	MIN. BURST (MPa - psi)	MAX. WORK (MPa - psi)	MIN. BEND (mm - in)
585-0026	10.1 - .75	22.2 - 1.1	12	10.0 - 580	10.0 - 580	240 - 9.45

Heavy Duty Straight Thread O-Ring - Hex Head Plugs

Part Number	Thread Size (in - FR)	O-Ring Seal
388-3887	1.5/16-12	1M 5465

STOR Port / STOR Port / Male STOR - Tee

Part Number	Thread size	L1 Length	Thread size	L2 Length	Thread size	L3 Length	Material
448-9011	1.5/16-12	33.9	1.5/16-12	33.9	1.5/16-12	57.15	Steel

Male ORFS / Male STOR

Part Number	Thread Size (A)	Thread Size (B)	HEX Size	Length (L)	Material
88-0197	1.5/16-12	1.5/16-12	38.1	83.5	Steel

ORFS Female Swivel-90° Elbow

Part Number	HOSE I.D. (mm - in)	HOSE O.D. (mm - in)	SEWER	THREAD SIZE	A LENGTH (mm - in)	B LENGTH (mm - in)	C FLARE OUT (mm - in)	D FLARE IN (mm - in)	E RING SEAL
124-2153	19.1 - 7/8	13	5/8"	1.316-12	12	64.3-5.31	52.8-2.07	47.6-1.88	65-5/16

Male ORFS / Male BSPP STOR-JIS B2351

Part Number	Thread Size (A)	Thread Size (B)	HEX Size	Length (L)	Material
88-0197	1.5/16-12	1.5/16-12	38	45	Steel

Parts List for Figure 5:

1	448-9011	T" Adapter
1	28-7001	Register
1	360-3697	Plug
4	239-6284	Seal
1	87-0197	Reduction
1	124-2153	Hose Terminal
1	228-7052	Seal
1	6V-9746	Seal
1	086-2746	Reduction
1	6V-0400	Seal
1	228-7052	Seal
1	124-2153	Hose Terminal
1	6V-9746	Seal

Picture 5 – Installation guideline to step 2

The on-board filtration system actuates only when the engine is functioning.

Necessary parts for fixing the hydraulic lines:

Below are the parts necessary to assist in the correct routing of the hydraulic lines, avoiding friction between the lines existing in the equipment.

Type	Pn	Qty	Description
Fix Hydraulic Lines	80-3174	2	Strimmel
Fix Hydraulic Lines	70-3834	2	Clamp A
Fix Hydraulic Lines	70-3833	2	Clamp B
Fix Hydraulic Lines	2005345	6	Support (Swivel Mount)
Fix Hydraulic Lines	1833973	12	Clamp 10.8" Width and 10" Length
Fix Hydraulic Lines	1772216	7	Clamp Dual Cable Tie Mount - 13" length
Fix Hydraulic Lines	149-7738	6	Clamps
Fix Hydraulic Lines	87-3426	1	Channel - 8mm (Rubber Protection)

Double Line Hose Clamps

Part Number	Part Description	Clamp Part A	Clamp Part B	Clamp Part C	Clamp Part D	Clamp Part E	Clamp Part F	Clamp Part G	Clamp Part H	Clamp Part I	Clamp Part J	Clamp Part K	Clamp Part L	Clamp Part M	Clamp Part N	Clamp Part O	Clamp Part P	Clamp Part Q	Clamp Part R	Clamp Part S	Clamp Part T	Clamp Part U	Clamp Part V	Clamp Part W	Clamp Part X	Clamp Part Y	Clamp Part Z
182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm	182 x 20.0 mm



Cable Tie Mounts and Straps

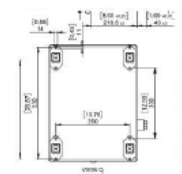
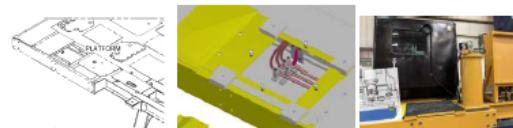
Part Number	Description
182-000	Strap Mount (Used for lines to hose at front of pump assembly)
182-000	Dual Cable Tie Mount - 13" length (Used to guide two hoses in parallel and prevent abrasion)
182-000	Cable Tie 13" Width and 13" Length

CIC Project - 794AC 10 of 18

Fixing the to the platform:

Parts for fixing the CIC filter station on the equipment platform. The rails are used to measure 80 centimeters and specific nuts (square) for fixing to the rail.

Type	Pn	Qty	Description
Fixing CIC	593-1627	2	Rails (2 meters each rail) Cut two pieces 80 cm.
Fixing CIC	05-1568	8	Screws M10x1.5x30-MMM
Fixing CIC	434-5654	8	Washer (M10x1.5-THD)
Fixing CIC	5M-2894	8	Washer 1.5x2x2.5-MMM THK



*** Rails (593-1627) were used to fix the CIC filter station. Cut two pieces to the size of 80 centimeters. Weld, as measured, the ends of the rails to the structure. Fix the rail to the structural plate of the platform.

Pictures 6 – Installation guideline to step 2

It should be noted that the 794AC's in Brazil do not have the hydraulic cabinets. The filtration unit was therefore installed in the location where the hydraulic cabinet would have been and next to the grease tank. This location provides good access for maintenance and inspection for both the filtration unit and grease tank (grease level).

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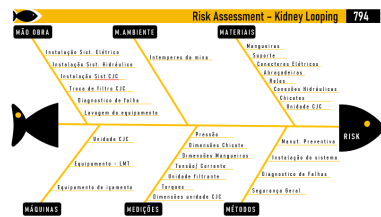
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Step 3: Training of the technicians on both operation and maintenance.

Training of the technician on both the operations and maintenance of the on-board filtration system was conducted through use of the documentation and in practice. All the safety aspects were identified, mapped, and mitigated during a risk assessment.



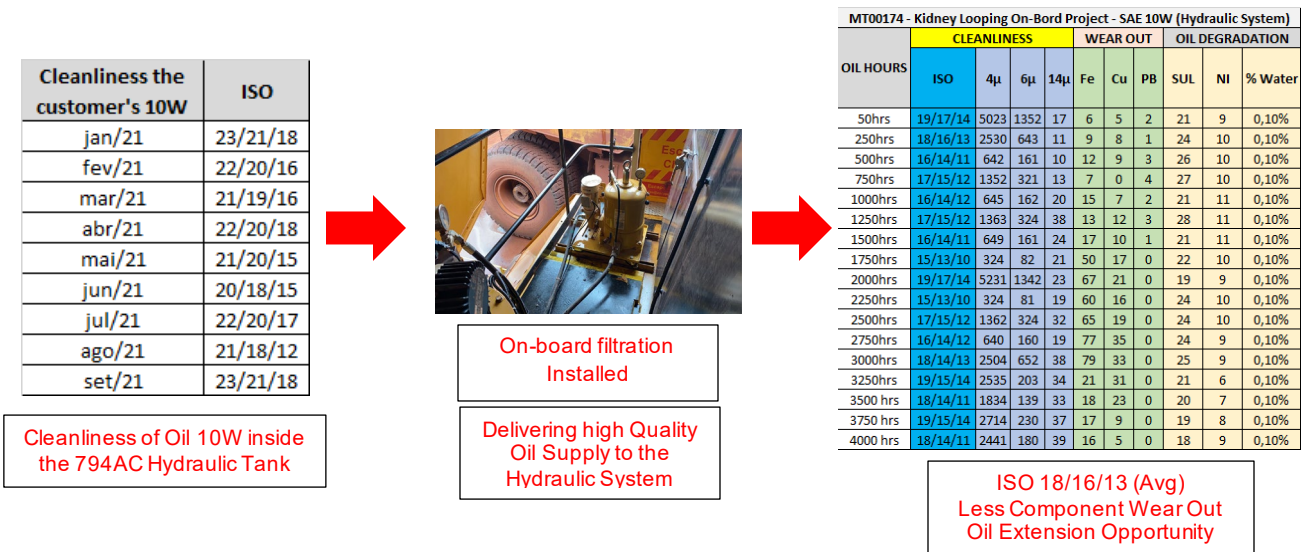
Pictures 7 – Service Training, On-board Filtration Guideline, Risk Assessment, Unit at Upper Platform

4.0 Benefits

1. Improved Cleaning of the Hydraulic System:

The on-board filtration system has proven effective in delivering results even with leaks and other existing contamination onboard as well as the high particle count (ISO > 18/16/13) in 10W oil in the customer's oil tanks that are used to fill the 794AC's.

Results for one of the trucks at 4,000 hours (Case study MT500174) are shown below.

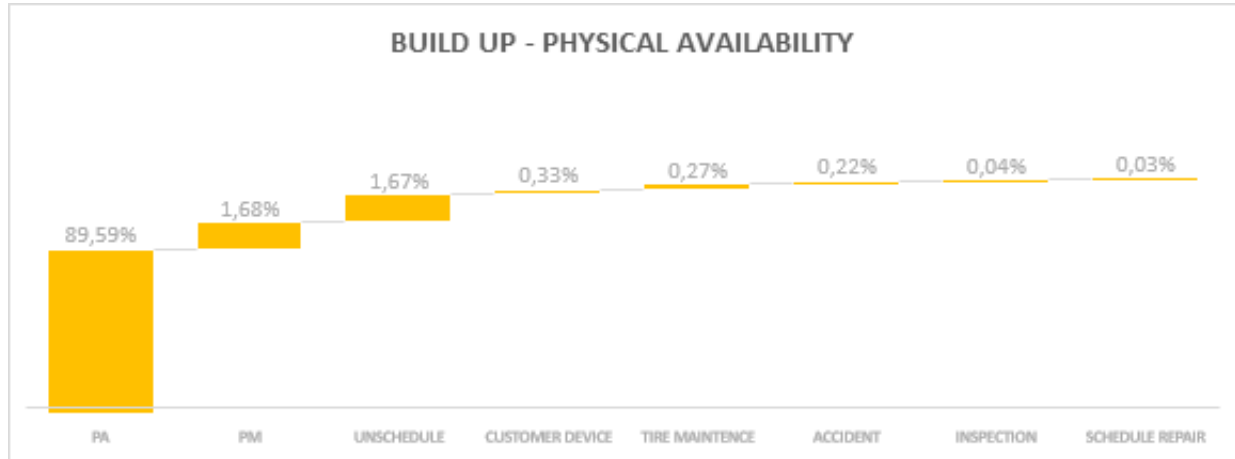


Case Study (MT500174) – Delivering better bottom line about cleanliness

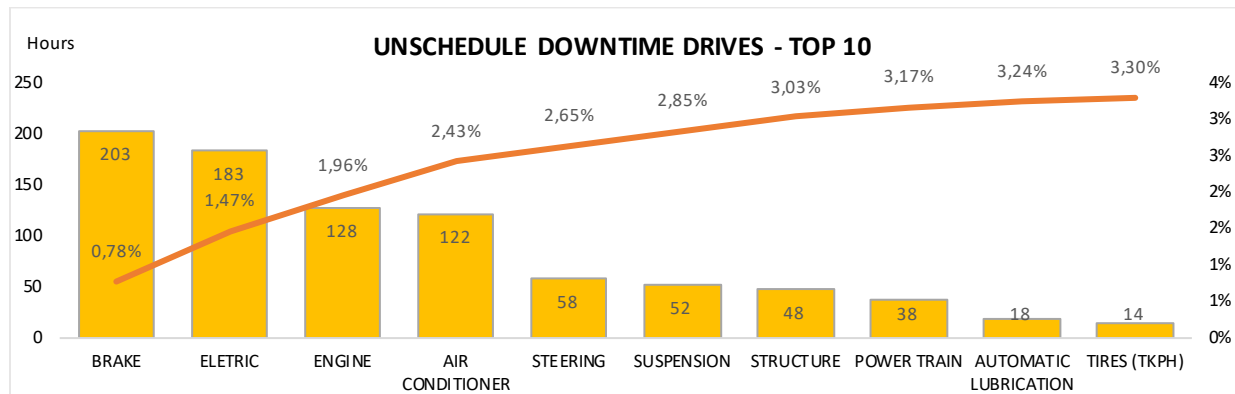
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2. Improved Reliability and Performance:

No exchange events in the hydraulic system, minor components and PTO were encountered between 0 and 4,000 hours. The onboard filtration system has contributed to a reduction in unscheduled downtime, even with outstanding CPI issues that have caused wear elements in the oil, or other downtime events. Ultimately, we expect to see major components meeting/exceeding life expectancy with the addition of this on-board filtration system.



Downtime Drives Accumulated (4,000 hours) – Hydraulic System had zero downtime events.

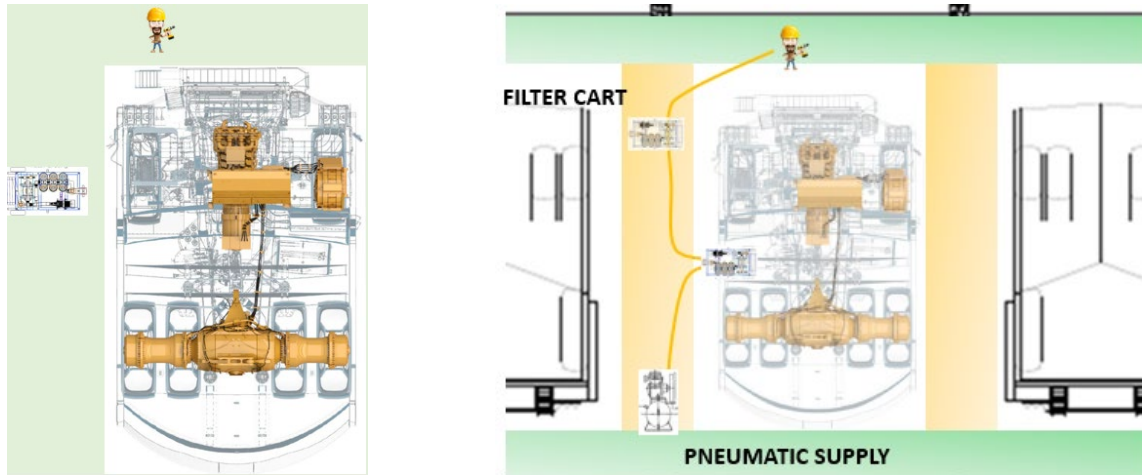


Downtime Drives Breakdown (4,000 hours) – No hydraulic system issues. The Brake System in this case was CPI failures about the piston wear out, not around the oil condition.

3. Improved PM Productivity:

Sotreq's execution team conducted an evaluation of the time and actual task routes/movements of the lubrication process, including fluid cleaning and supply of the trucks. Use of the on-board filtration system instead of the off-board kidney looping machine has eliminated 6 hours and 20 minutes of work from the PM process. Additionally, this system has both increased the PM productivity and reduced the maintenance ratio of the machine.

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Flow Diagram – Off-board kidney looping process

Activities	Men Hour	Duration
Safety Documents	00:25	00:25
Preparation	00:15	00:15
Filter Cart Instalation	00:30	00:30
Filtration*	00:00	04:00
Particle count	00:25	00:25
Filter Cart Removing	00:30	00:30
Cleaning and Storing	00:15	00:15
Total	02:20	06:20

Chart 1 –

Approximately 6 hours of manpower can be removed from the PM process by replacing the off-board kidney loop process with the new on-board process.

*Average time to perform the filtration when ISO is above of limits.

5.0 Resources Required

Project Investment Cost – Please contact the Project Sponsor

6.0 Supporting Attachments / References

Contamination Control Guideline, Customer Feedback Survey, 794AC OMM.

7.0 Related Best Practices

A PM Model do deliver higher availability of 794AC fleet – 1.01.210827-1

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